

POLS0008

INTRODUCTION TO QUANTITATIVE RESEARCH METHODS

WEEK FOUR: SOURCING DATA

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QUICK RECAP OF WEEK 3

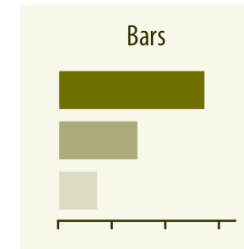
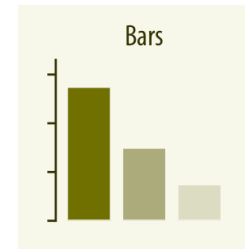
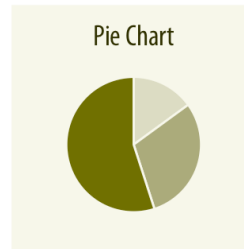
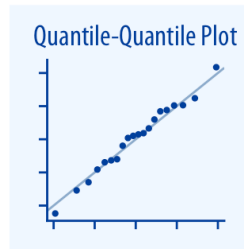
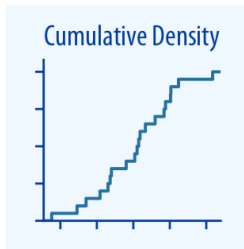
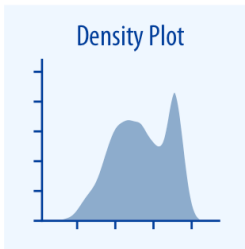
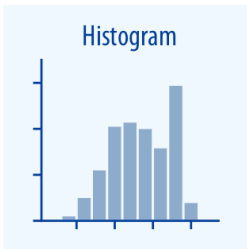
We covered some general techniques for data visualisation

Visualisations representing densities & distributions of numerical data

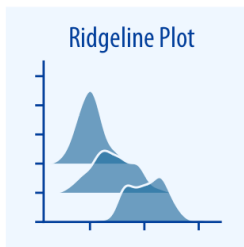
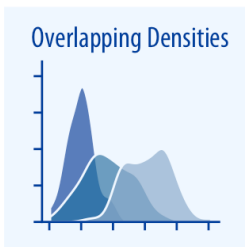
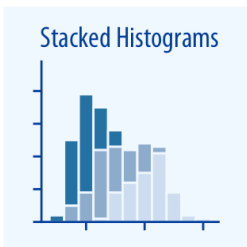
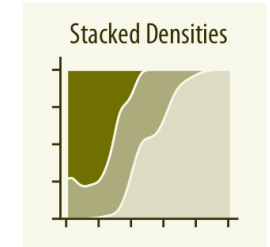
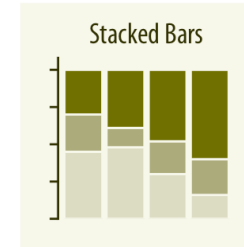
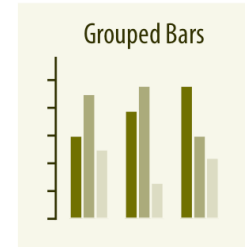
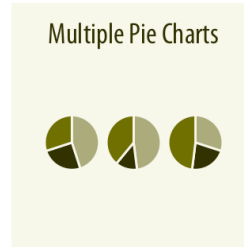
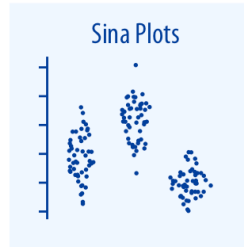
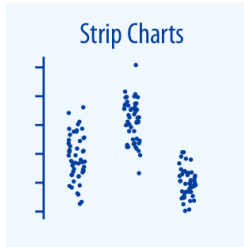
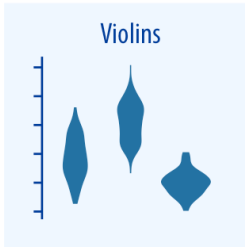
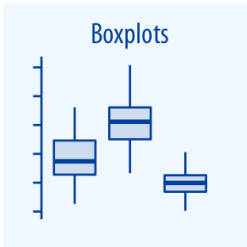
1. Plots for densities and distributions (numerical data)

2. Plots for proportions (qualitative or categorical data)

Single variable



Multiple variables

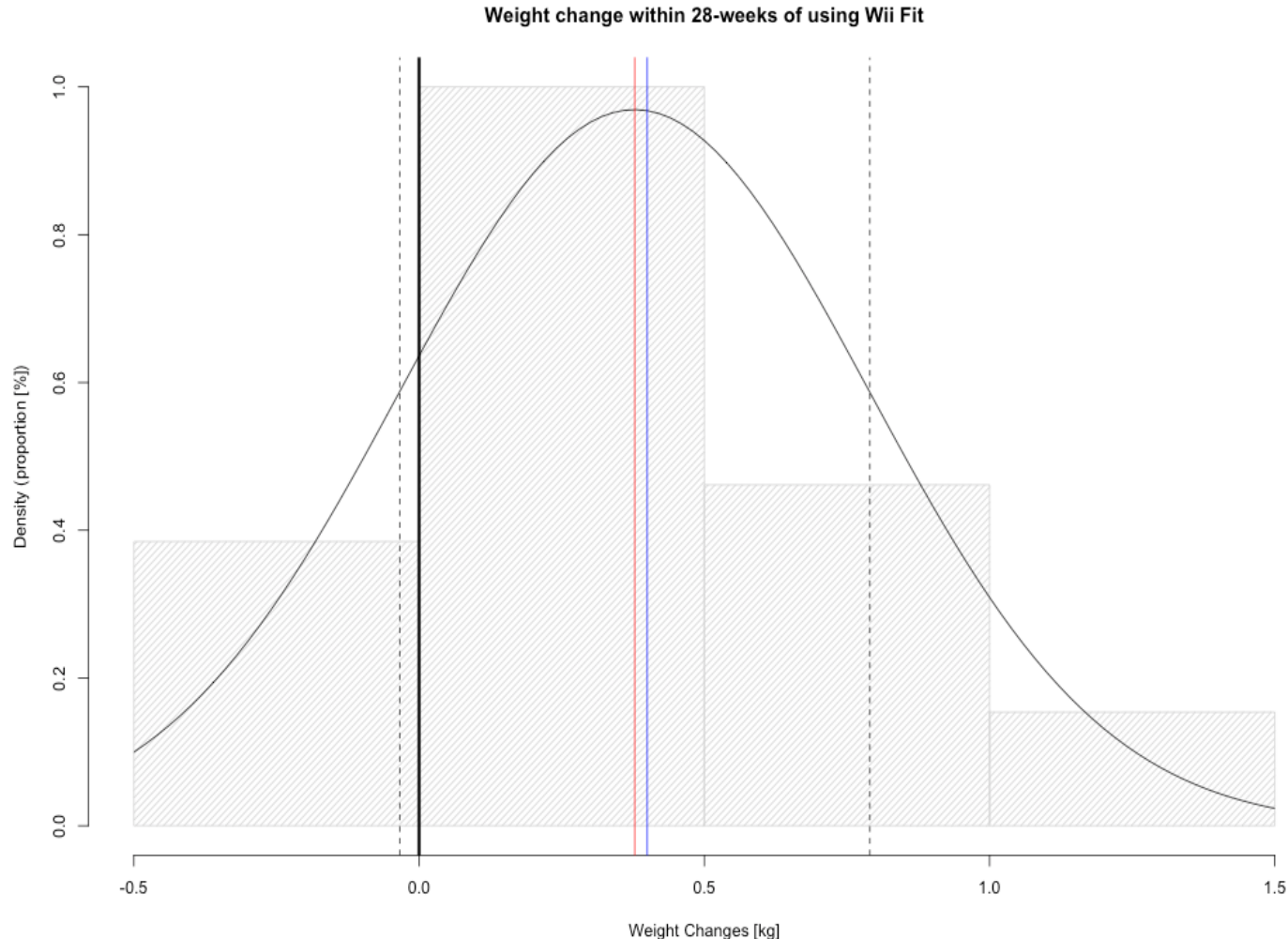


We covered the best practices for data visualisation

Everything on your graph should be labelled accordingly:

- **Title** – a clear short title letting the reader know what they are looking at should be present. Or alternatively, a figure legend for that image will do.
- **Axis labels/titles** – clear labels for the x and y axes must be present
 - ❖ Should include in the labelling the units of measurements [height (m), soil arsenic (mg/kg) etc.]
 - ❖ These labels should be short and descriptive
- **Legends** – for categories in categorical variables which keys/colour codes must be present and labelled accordingly
 - ❖ Male and Female, and not 0 and 1.
- **Captions** – If the graphics are **NOT** yours (i.e., its ripped from a source). Take the opportunity to apply a caption on the graph (on or beneath it) providing source attribution for the data.
- **Colour scheme** – Use of colour scheme matters
 - ❖ Sequential colours – for plotting quantitative variable that goes from low to high (vice versa)
 - ❖ Diverging – for contrasting the extremes (low, medium and high) of a quantitative variable
 - ❖ Qualitative – e.g., nominal categories. Use to distinguish between different categories in a categorical variable

We covered an example using Base-R code



Note: Area beneath the curve before weight change value at 0 kg (black solid line), tells me the predicted probability that the Wii Fit can reduce my weight.

$\text{Prob}(\text{Weight change} < 0.0\text{kg}) = 0.1794$ (17%)

R Code:

```
# enter data
weightChanges <- c(-0.2,-0.4, 0, 0.1, 0.1, 0.3, 0.12, 0.4, 0.5, 0.8, 0.9,
0.5, 0.6, 0.7, 0.6, 1.3, -0.2, 0, 0.1, 0.1, 0.3, 0.4, 0.5, 0.5, 0.6, 1.2)
```

```
# next, extract mean and standard deviation from data
m<-mean(weightChanges)
std<-sd(weightChanges)
```

```
# plot histogram with normal curve
hist(weightChanges, density = 20, prob=TRUE, main="Weight change within
28-weeks of using Wii Fit", xlab = "Weight Changes [kg]", ylab = "Density
(proportion [%])")
```

```
# adds the normal curve
curve(dnorm(x, mean=m, sd=std), add=TRUE)
```

```
# add red line for mean
abline(v = 0.378, col = "red")
```

```
# add blue line for median
abline(v = 0.400, col = "blue")
```

```
# add black dashed line for -sd
abline(v = -0.034, lty = "dashed", col = "black")
```

```
# add black dashed line for +sd
abline(v = 0.79, lty = "dashed", col = "black")
```

```
# add blue line for median
abline(v = 0, col = "black", lwd = 3)
```

```
# Calculate probability
pnorm(0, mean=m, sd=std)
```

We demonstrated further ways of visualising data in RStudio with ggplot()

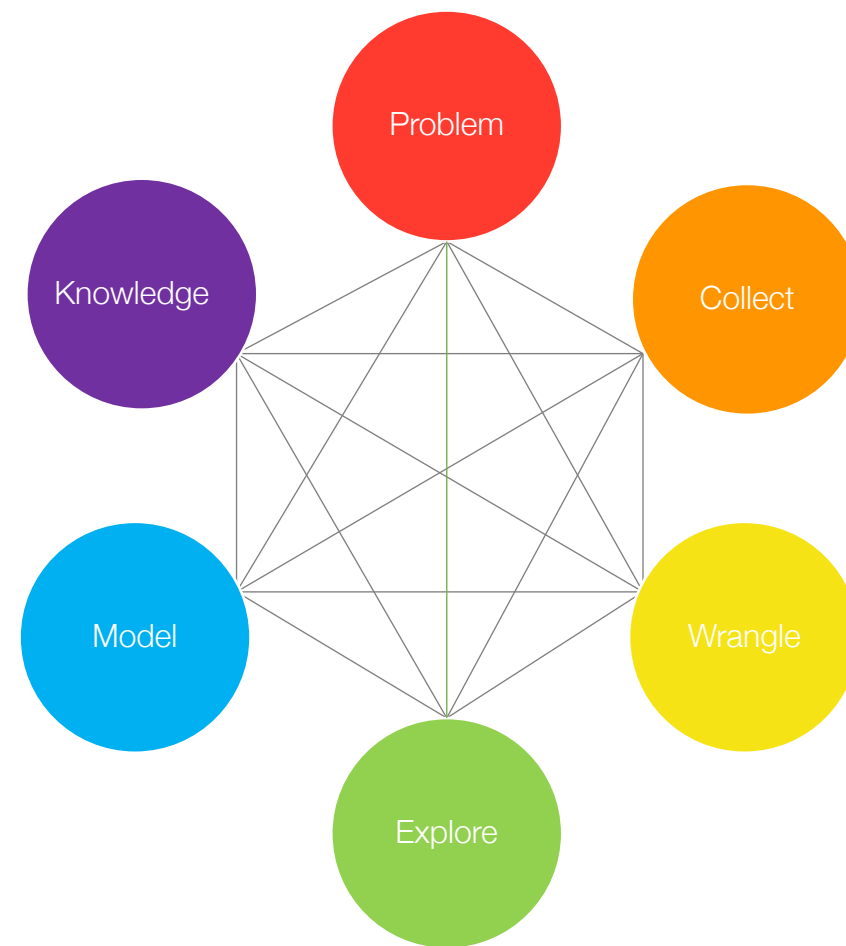
Creating impressive visualisation

- Base functions for creating graphics in R: `plot( )`
- R Packages for creating impressive plots: `ggplot2( )`
 - You will have to first install 'ggplot2' package first with the `install.package( )` function
 - After its installed, you will need to load the package into R with `library( )` function
- All this will be become clear in the practical session

Data Sourcing

Contents

- What's data sourcing?
- The distinction of primary and secondary data sources
- Examples of sourcing data
- Summarising the merits and limitations of using primary or secondary data
- Brief notes on study design and sampling strategy
- Description of data



Format of today's lesson goes...

Theory & Application

Student Feedback: 5-10 minutes

Break: 15 minutes

Description of Data

Summary of weeks 1-4

Lecture

What is Data Sourcing?

- Data sourcing (or data collection) is referred to how one goes out to gather data – this can be done either ‘ACTIVELY’ or ‘PASSIVELY’
- ACTIVE: these are ways of acquiring records (or data) actively through fieldwork, surveys, interviews etc.,
- PASSIVE: these are ways of acquiring records (or data) that are made available from some “official” (or “recognised”) sources (e.g., NGOs, agencies, companies, government, educational or research institutions etc.,)
- Official (or recognised) sources can make their data freely available to everyone to use & republish, repurpose it as they wish without any patent, legal or copyright restrictions, or through a secured digital channel (e.g., Data Safehaven)
- On the fourth point, we often refer to this term as “Open Data” [\[LINK\]](#) for datasets that are freely available. While, the term “Safe-Guarded Data” is used for those acquired through secured channels.
- Open & Safe-Guarded Data are passive & falls under the broad type of data classification called “Secondary Data”, whereas the second point on active data collection is called “Primary Data”

Types of data sources

Primary data

This type of data source refers to the firsthand data gathered by the user, researcher or enumerator through fieldwork, interviews, questionnaire surveys etc.,

Secondary data

This type of data source typically refers to the data that has already been collected through a primary source and made readily available for other researcher(s) to use for their study or investigation

Internal	External
----------	----------

Secondary data

This type of data source typically refers to the data that has already been collected through a primary source and made readily available for other researcher(s) to use for their study or investigation

Internal Secondary data

- If you are working within an organisation that has some relevant data of interest
- The organisation has collected the data and can provide access to such users working within it or collaborators
- Data scientists working within the UK Metropolitan Police Service seeking to quantify the burden of various crime outcomes: Burglary, sexual assault, vandalism, arson and so...
 - Data is collated by MPS and not by the data scientists
 - So MPS can release the data to such users, this is an example of **Internal Secondary Data**

Secondary data

This type of data source typically refers to the data that has already been collected through a primary source and made readily available for other researcher(s) to use for their study or investigation

External Secondary data

- This simply refers to acquiring open datasets from external sources
 - Open-source websites (freely accessible)
 - Paying for the data (requires a license)
 - Online data service which is free but requires users to register etc.,
 - Release for FOI (Freedom of Information)

Sometimes you can combine both internal and multiple external data sources to supplement each other!

The umbrella term 'secondary data' can also be referred to as "routinely collected data"

Examples of Sourcing Data

Example of primary data: Development Frontiers in Crime, Livelihoods and Urban Poverty in Nigeria (FCLP) [1]

Themes:

1. Criminology
2. Social Science
3. Qualitative Research
4. Quantitative Research
5. Global South



Conventional analyses of crime, based on European research models, are often poorly suited to assessing the specific dimensions of criminality in Africa. Development Frontiers in Crime, Livelihoods and Urban Poverty in Nigeria (FCLP) aims to provide an alternative framework for understanding the specific drivers of criminality in a West African urban context.

Employing a mixed-methods approach combining statistical modelling, geo-visualisation and ethnography, the project situates insecurity and crime against a broader backdrop of rapid urban growth, seasonal migration, youth unemployment and informality. The study provides researchers both in Nigeria and internationally with a richer and more nuanced evidence base on the particular dynamics of crime in African cities.

Example of primary data: Development Frontiers in Crime, Livelihoods and Urban Poverty in Nigeria (FCLP)

See source(s):

1. [Musah et al, 2020](#)
2. [Umar et al, 2020](#)
3. [Umar et al, 2017](#)



Data collected:

- Block Inventory Survey for the collection of environmental data
- Household victimisation survey on indicators for crime
- Perception of risk and neighbourhood safety
- Demographic survey

Primary data and sampling strategy: 13,000 households, and the target sample was 2,300 for the victimisation survey (in Nigeria); we therefore used Systematic sampling to select at random 2,300 households [the applied criteria: $k = 13,000/2,300 = 5.6 \sim 6^{\text{th}}$ property (starting from the left-side of road)]

Houses mapped in B, we had to interview all those residents (i.e., 2,300 households) to collect a range of information.

Physical characteristics variables

BLOCK ENVIRONMENTAL INVENTORY FORM		Block Ref.	
Road	Type:	Surveillance	CCTV Camera:
	Condition:		
Traffic	Vehicular	Morning	Afternoon
	Pedestrian		
s/n	Ref No	Land use	Plot
1	2	3	4
Occupied	vacant	Abandoned	Construction
Shops	Kiosk	In-trading	Out-trading
Flat	Storey	Attached	Detached
Cement	Mud	Temporary	High walls
Burglary-proof	Gate	Garage	Outdoor sitting
Security Lights	Guards	Open Access	Danger Warning
CCTV Camera	Dogs		

Section A: Questions related to household Details

1. Are you the household head? Yes [] No []
If No, please indicate your relationship to the household head _____
2. a) Sex: Male [] Female []
b) Age: []
c) Ethnicity: _____
3. Occupation: Civil Service [] Private Organisation [] Craftsman [] Trader []
Farmer [] Student [] Retiree [] Unable to work [] Unemployed []
Others, please specify _____
4. Employment Level: Executive [] Managerial [] Expert [] Intermediate []
Trainee [] Large business proprietor [] Small business proprietor [] Others,
please specify _____

+

Section B: Questions related to your relationship with neighbours and your perception of safety within this street

Note: - Properties in a street are those on both street block faces between two road intersections
- Neighbours are those people who live in the same street with you

1. How safe do you feel living on this street? Extremely safe [] Very safe []
Moderately safe [] Slightly safe [] Not safe at all []
2. How worried are you about being a target of property crime while you are away from home?
Not worried at all [] Slightly worried [] Moderately worried []
Very worried [] Extremely worried []
3. How many of your neighbours do you know? All of them [] Most of them []
Half of them [] A few of them [] None of them []

+

Section C: Questions related to incidents that had happened within your property

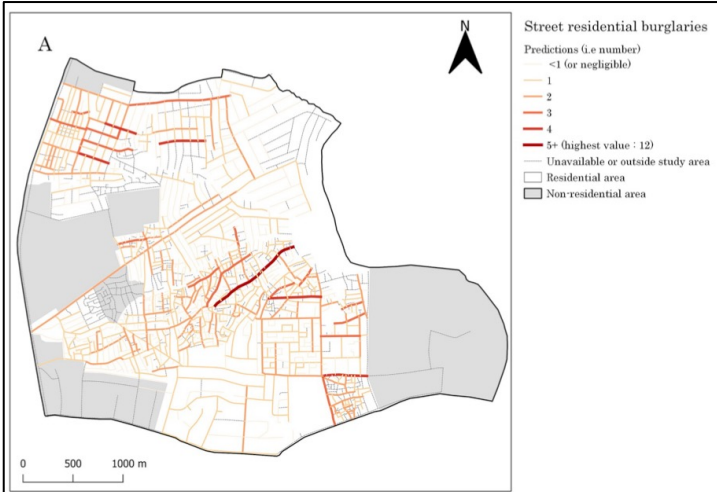
In the **LAST 1 YEAR**, have any of the following incidents **HAPPENED** within your Property?

1. Burglary (Breaking-in) - Yes [] No []
If yes, how many times? []
2. Stealing of valuables (Not breaking-in) - Yes [] No []
If yes, how many times? []
3. Deliberate damaging of your property Yes [] No []
If yes, how many times? []
4. Theft **from** Automobile Yes [] No []
If yes, how many times? []

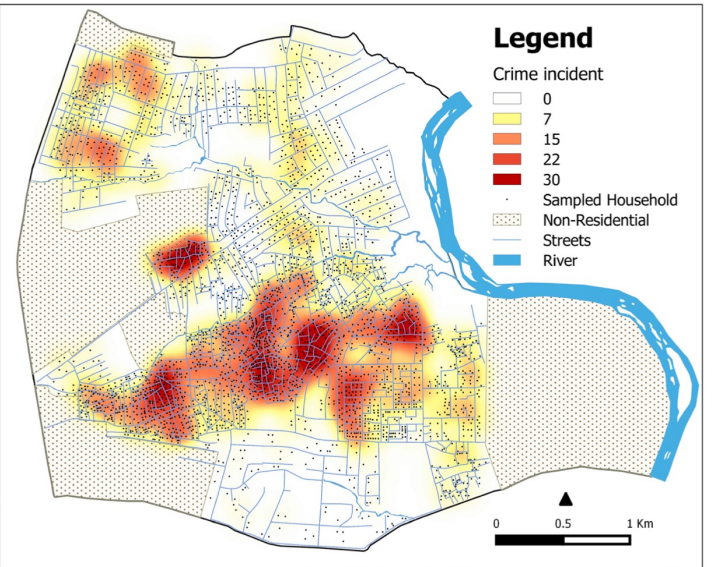
Sociodemographic variables

Perception and safety variables

Victimisation (dependent) variables



Research 1: From the 2,300 household sample, we used the “crime pattern theory” to assess the risk of burglaries & victimisation at a street-level (see source: [LINKI](#))

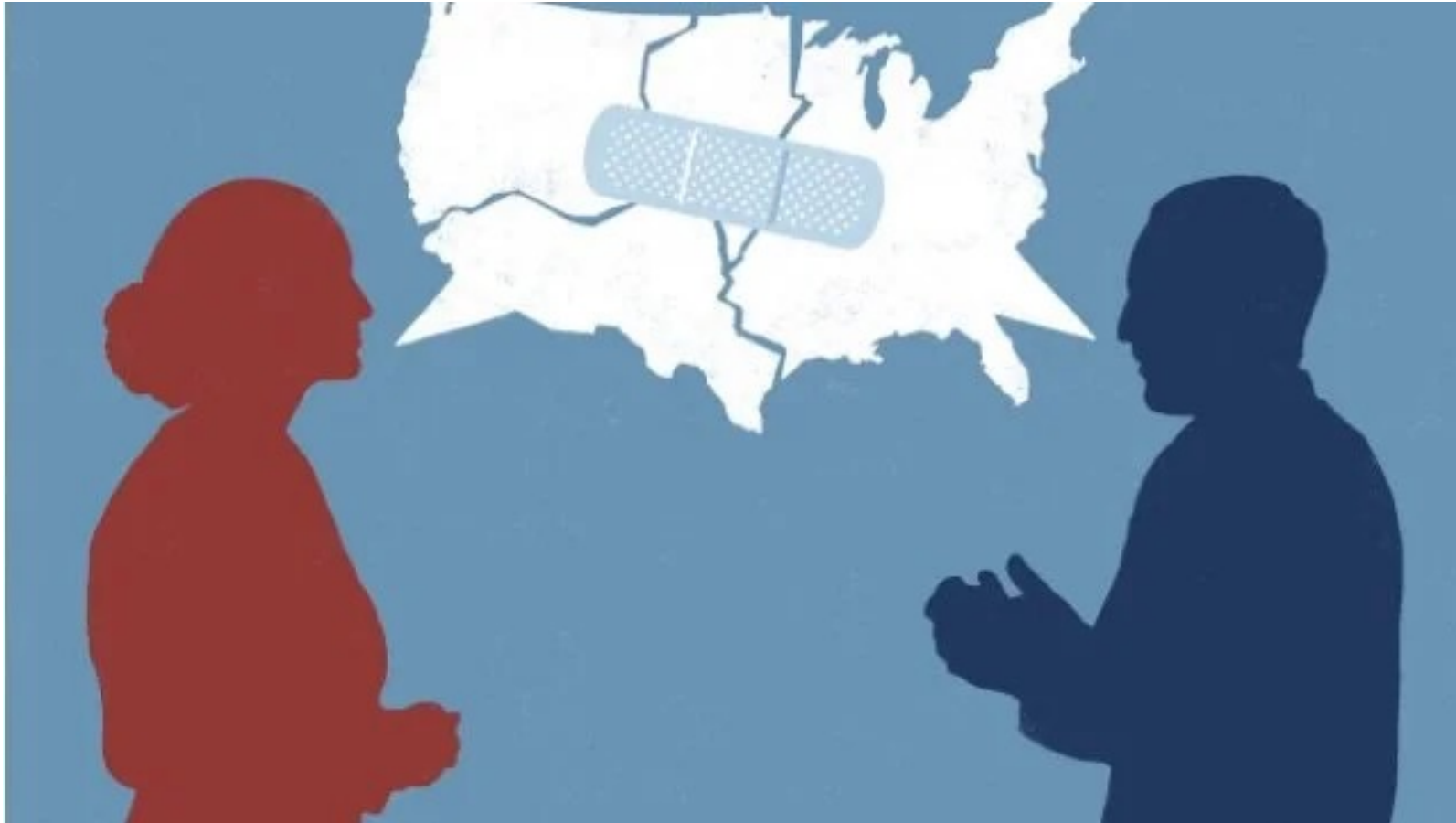


Research 2: From the 2,300 households were sampled, we used the “laws of crime concentration” to assess the concentration of reported victimisation in this city (see source: [LINKI](#))

Example of secondary data: The geography of partisan homophily in the United States

Themes:

1. Social Science
2. Geography
3. Quantitative Research
4. Politics
5. Social Media



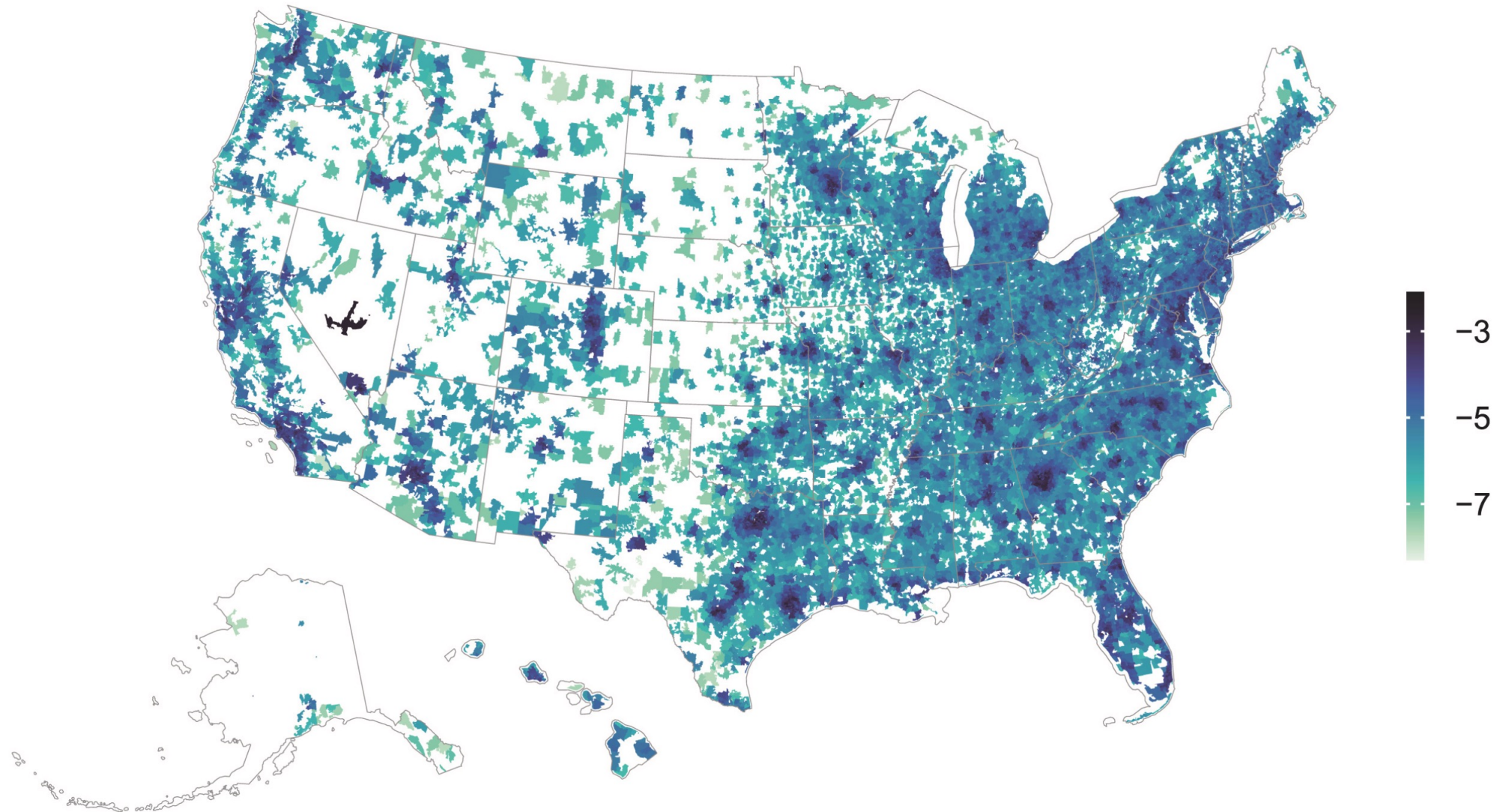
[Image source](#)

Political homophily in the United States is often proxied by residential proximity. The assumption is that if most people in a given neighbourhood vote Republican (or Democrat) then most of these people's friends will also vote Republican (or Democrat). But as residential proximity is an imperfect proxy, we do not truly know the extent of political homophily across the US.

This study has created the first detailed country-wide map of political homophily in the United States based on actual data on the social ties among people. Combining Facebook data on aggregate social media friendships between US zip codes, and voting records from the 2020 US presidential election it shows which neighbourhoods in the United States are more likely to be socially connected to politically similar others.

Example of secondary data: The geography of partisan homophily in the United States

A. Log. of relative connectedness to social neighbours



Facebook data:

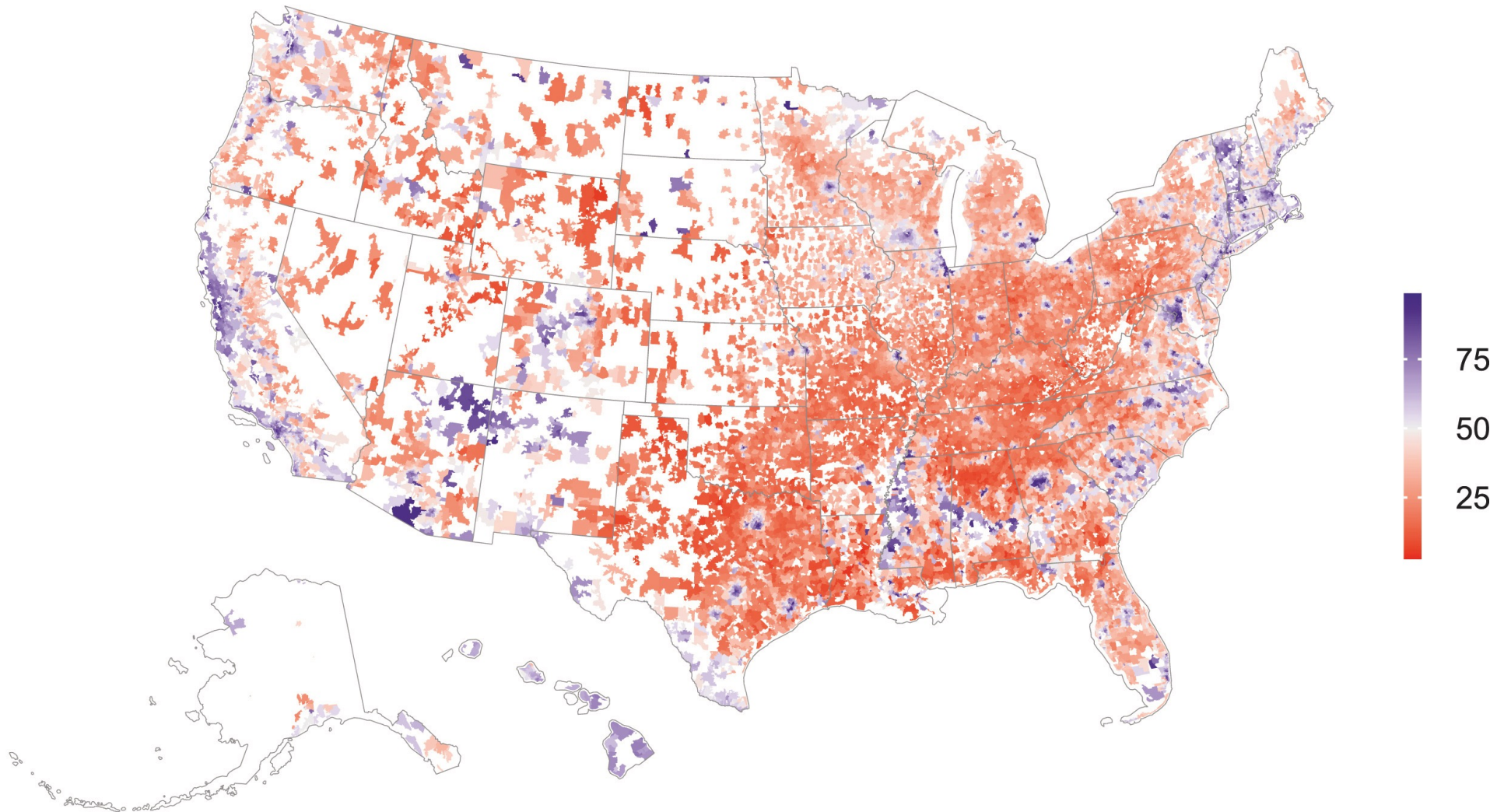
- 'Social Connectedness Index' produced by Meta
- [Meta Datasets](#)
- Measures density of Facebook friendships between zip codes
- 22,537 zip codes covered

Source:

[Mastrosawas, A. \(2024\)](#)

Example of secondary data: The geography of partisan homophily in the United States

Democratic two-party vote share in the 2020 US presidential election, by zip code



Voting records

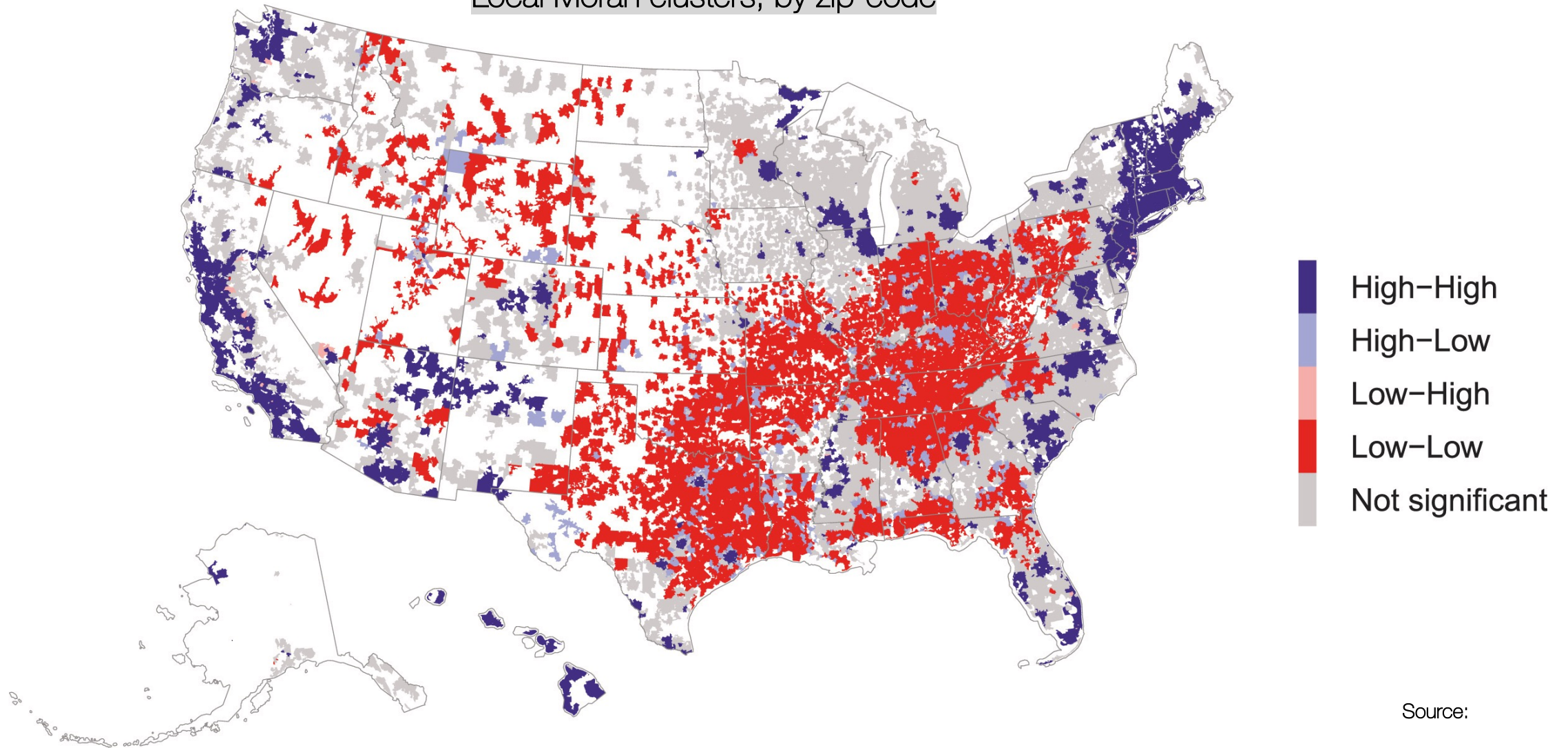
- Collected by local elections boards
- Actual voting behaviour in the 2020 US presidential election

Source:

[Mastrosawas, A. \(2024\)](#)

Example of secondary data: The geography of partisan homophily in the United States

Local Moran clusters, by zip-code



Source:

[Mastrosavvas, A. \(2024\)](#)

High-High: HIGH Democratic vote share in the area and HIGH in its socially connected areas
High-Low: HIGH Democratic vote share in the area and LOW in its socially connected areas (etc)

Examples of open primary data for health research in Global South



THE MALARIA ATLAS PROJECT



Example of secondary data: Consumer Data Research Centre (UCL)

The screenshot shows the CDRC website interface. At the top, there's a logo for the Consumer Data Research Centre, an ESRC Data Investment, and a search bar. Below the logo is a navigation bar with links to CDRC, Datasets, Stories, Tutorials, Topics, Geodata Packs, and About Data. There are also links for Log in and Register. The main content area is titled 'Home » Dataset » Search'. On the left, there's a sidebar with filters for Content Types (Dataset selected), Topics (Population & Mobility (42), Retail Futures (22), Finance & Economy (12), Transport & Movement (8), Digital (6)), Type (Open (37), Safeguarded (28), Secure (13)), Controller (University College London (UCL) (54), University of Liverpool (13), University of Leeds (11)), and Years. The main search area has a search bar, a 'Reset' button, and sorting options (Sort by: Relevance, Order: Descending, Apply). Below the search bar, there are 78 results. The first result is 'High Street Retailer - Retail and Consumer Data' (Secure), which is a Retail Futures dataset. The second result is 'Airbnb Property Rentals and Reviews (supplied by AirDNA)' (Safeguarded), which is also a Retail Futures dataset.

Consumer Data Research Centre
An ESRC Data Investment

search

CDRC Datasets Stories Tutorials Topics Geodata Packs About Data Log in Register

Home » Dataset » Search

Content Types
Dataset
Topics
Population & Mobility (42)
Retail Futures (22)
Finance & Economy (12)
Transport & Movement (8)
Digital (6)
Type
Open (37)
Safeguarded (28)
Secure (13)
Controller
University College London (UCL) (54)
University of Liverpool (13)
University of Leeds (11)
Years

Search
Search
Reset

Sort by
Relevance
Order
Descending
Apply

78 results

High Street Retailer - Retail and Consumer Data Secure
Retail Futures
The High Street Retailer datasets contain information on customer and retail characteristics and transactions from one specific high street retailer (only) which has a presence on many of the high streets of the UK as well as in some other...

Airbnb Property Rentals and Reviews (supplied by AirDNA) Safeguarded
Retail Futures
This data profile describes a dataset held by the CDRC which has been supplied by AirDNA LLC. AirDNA provides data and analytics to vacation rental entrepreneurs and

CDRC provides data for research to address societal and economic challenges in the UK.

Contains tonnes of open, safeguarded and secure data.

Website: <https://data.cdrc.ac.uk/search/type/dataset>

Search for a keyword(s) or time series ID



Coronavirus (COVID-19)

[Get the latest data and analysis on coronavirus \(COVID-19\) in the UK.](#)

Main figures – [From our time series explorer](#)

Employment

Employment rate

Aged 16 to 64
seasonally adjusted
(Sep - Nov 2021)

75.5%

↑ 0.5pp on previous
year

[Analysis](#) [Data](#)

Unemployment rate

Aged 16+ seasonally
adjusted (Sep - Nov
2021)

4.1%

↓ -1.0pp on previous
year

[Analysis](#) [Data](#)

Inflation

CPIH 12-month rate

Dec 2021

4.8%

↑ 0.2pp on previous
month

[Analysis](#) [Data](#)

GDP

Quarter on Quarter

Jul - Sep 2021

1.1%

↓ -4.3pp on previous
quarter

[Analysis](#) [Data](#)

ABOUT ONS:

It is responsible for the collation census data relating to the UK population, as well as the publication of statistics related to the UK economy, it's population and wide range of societal matter as a whole.

UK government makes use of these statistics for their policy and decision making.

Actual raw UK data can be found from ONS and UK GOV.



Consumer
Data
Research
Centre



Office for
National Statistics

census [\[LINK\]](#)
2021



GOV.UK

IMD: [\[LINK\]](#)



Ordnance Survey

LONDON DATASTORE

[\[LINK\]](#)

DATA.POLICE.UK

[\[LINK\]](#)

Merits & Limitations

Advantages and disadvantages of Primary data sources

Advantages

- Data collected is always up-to date
- Relevant and specific to user's research aims and objectives
- High-level, and greater of understanding about the nature and content of the dataset
- High-level of accuracy as long as you do whatever it is in your power to minimise all kinds of systematic errors in the data collection process (ensuring **Internal Validity**)

Disadvantages

- Depending on the study design – data collection is a very time consuming and expensive process
- If you are collecting personal and sensitive data, you must **apply** for ethical approval before going out to get your data
- You will have to clean, manage and maintain your own data
- Possibility to falsify his/her data since one has his/her autonomy over the data

Advantages and disadvantages of Secondary data sources [2]

Advantages

- Ease of access and low cost, or even free if the data is from an open-source platform.
- Time-saving, especially if the data has already been processed and cleaned
- Unlike primary data; secondary data are often collected routinely hence allowing for longitudinal analysis
- Often secondary data are combined with different sources making it have a variety of variables.

Disadvantages

- You have no control over the quality
- The secondary data might not be specific to your needs
- The data can be biased in favour of the one whose gathered it. User will not know of this data artefact!
- You are not the owner and will never have full understanding of its nature & how it was collected (i.e., its essentially a “Blackbox”)

Brief notes on Study Design & Sampling

Definition:

“Research Design typically refers to the investigator’s strategy or plan for tackling a research question through collection of data, analysis and interpretation of such data, and finally a thorough discussion of that said data.”

In other words, or simply put it in plain English:

“... it’s someone’s blueprint for answering a research question”

Types of Research Design

Quantitative

This area allows the researcher to derive meaningful insight (or empirical evidence) about certain phenomena through analysis of numerical information

- Descriptive studies
- Observational Studies
- Experimental Studies

Purpose: Evidence of causality (or an association);
Internal validity and External validity

Qualitative

This area allows the investigator to gain meaningful insight (or empirical evidence) of certain phenomena through the study of non-numerical pieces of information

Types of Study Design

Type	Study Design	Properties
Descriptive	• Ecological (or geographic) • Cross-sectional	• Unit of observation are at group- or aggregated level (e.g., geographic unit) • Data is collected on a single snap in time, and its useful for generating further hypothesis
Observational	• Case-control • Cohort (or longitudinal)	• Observe the effect of an independent variable(s) on an outcome that has already occurred. The time frame is a retrospective analysis • Observe the effect of an independent variable(s) on an outcome that has not happened yet. The time frame is a prospective analysis
Experimental	• Randomised control trials (RCT)*	• Observe the effect in an actual test group(s) (e.g., intervention, effectiveness of a teaching programme, clinical trials for a vaccine)

Pilot study is a special cases as it could be anyone of these study designs

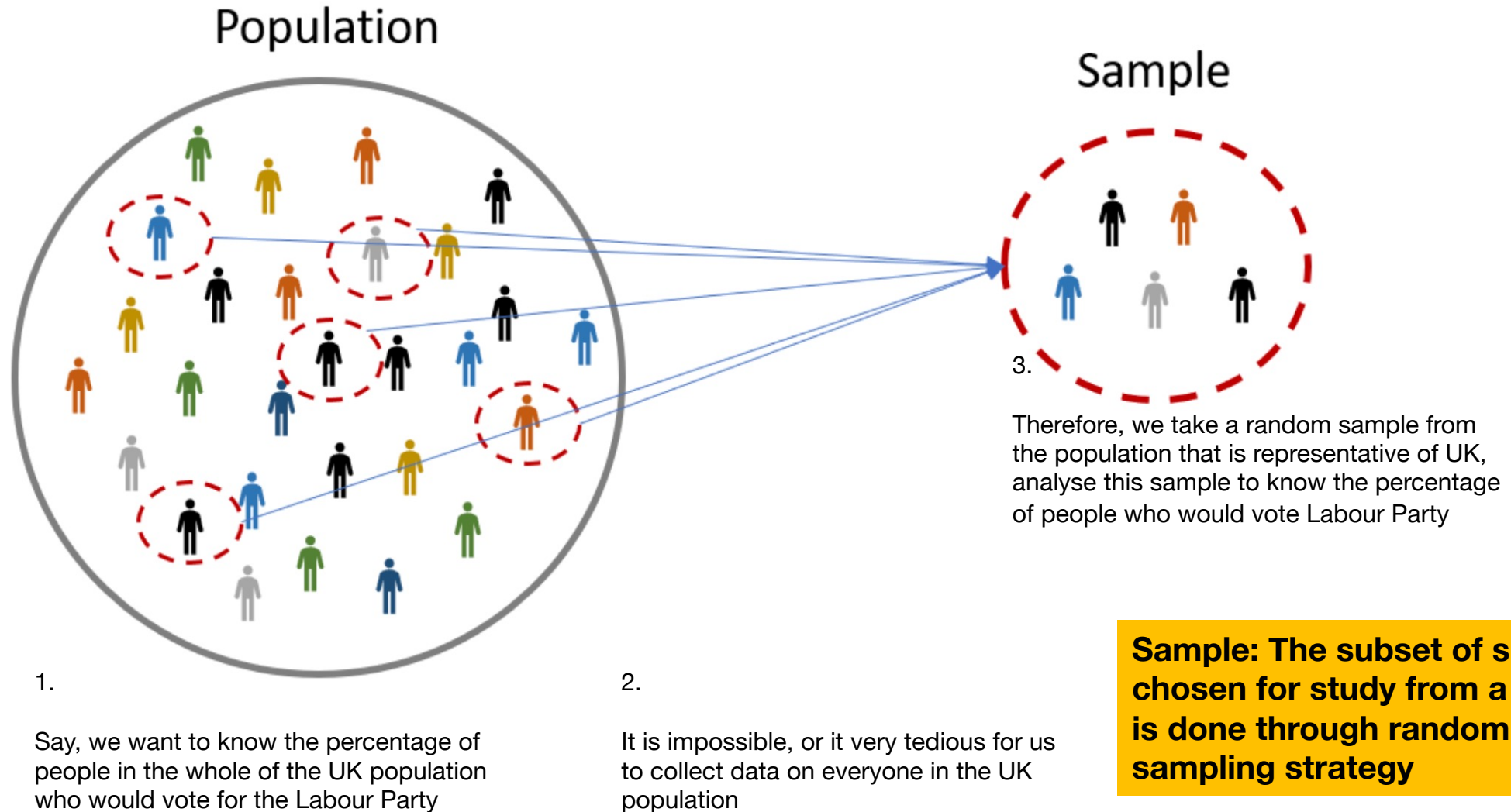
Comparison of quantitative-based study designs

	Ecological study	Cross-sectional	Case-control	Longitudinal	Pilot
Resolution of study design	This solely deals with aggregated units of data for analysis	Individual-level at a particular point in time	Individual-level where the outcome has already been observed among cases. There must be a control population	Individual-level at several points in time	Could be either aggregated or individual-level
Unit of time analysis	Has the flexibility of being a cross-sectional (i.e. single point in time) or longitudinal (i.e. several different time points).	At a single point in time	Past exposure	Several points in time, or before/after	Has the flexibility of being either ecological, cross-sectional or longitudinal study but dealing with smaller sample size as pilot before to doing a much bigger study.
Its cost effectiveness	Cheapest as it relies on routinely collected data most of the time	Less expensive as it conducted as a single time point and requires less resources	Quite expensive to interview participants to provide past experiences	Most expensive as it requires two or more follow-up of subjects enrolled in the study so more resource and time are required.	It's a cheap way for assessing whether to do a bigger (e.g. population-based) study
Common biases (or limitation)	Ecological fallacy	Results are only representative at time of study	Recall Bias	People dropping out of study can introduce lost-to-follow-up bias	Safest options as it's a pilot
Strength	Weakest	Strong	Strong	Strongest	Safest option
Retrospective or prospective?	Both	Present or retrospective	Always retrospective	Always prospective (even it's using historical data)	Both

Sampling Strategy [1]: Recall the vote intention example in Week 1?

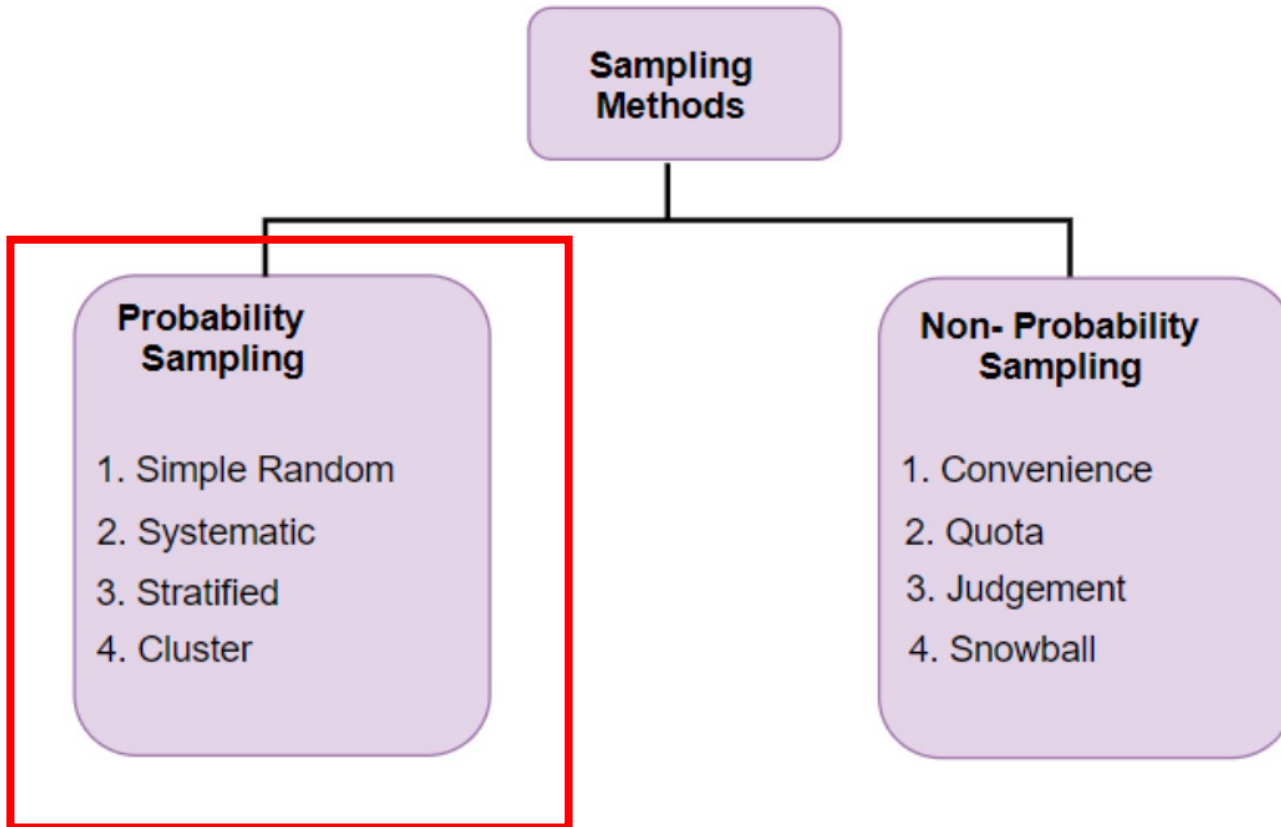
Parameter: The proportion of people in the UK population who would vote for the Labour Party (p)

Statistic: The sample proportion of people from the UK population would vote for the Labour Party ($\hat{p}$)



Sampling Strategy [2]

Sampling Method



Notes:

- **Simple Random Sample** is a subset of population in which each member of the subset has an equal chance of being select at random.
- **Systematic Random sampling** is another probability sampling approach where you have a list of individuals from a target population and perform a selection by starting somewhere at random and then collecting data at a fixed interval from that starting point.
- **Stratified Random Sampling** is an approach that requires one to divide the population and group them by a common characteristic, and then start sampling individuals from each groups at random
- **Cluster Sampling** is an approach that requires one to select a cluster unit(s) at random and then survey everyone in the selected cluster unit (usually smaller populations such a villages, or small schools in remote areas etc.,)

Warning: You cannot use any statistical method on samples that were collected with non-probabilistic approach – that element of randomness has been removed and thus render the sample not representative of the population. If you intend to use the quantitative route do not use a non-probability sampling approach.

Break

Description of data

Description of Data [1]

BEFORE YOU START THE ANALYSIS - you are required to provide some information about the dataset you are using regardless if its primary/secondary

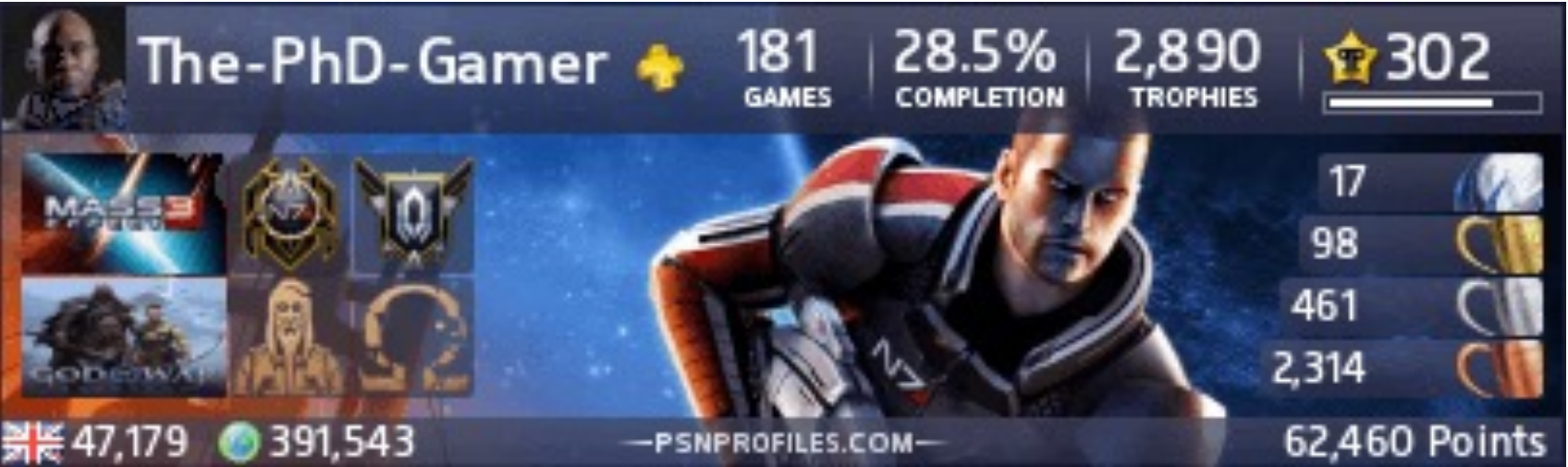
- **Describing the where, what and when:**
 - ❖ **What:** Here, you are mentioning the name of the data source and its use for a bigger study
 - ❖ **When:** The year(s) (or date(s)) at which it was collated
 - ❖ **Where:** The location of focus for which the data was collated from
- **Specific details about how the data was collated:**
 - ❖ Whether its through questionnaire survey, interview etc.,
 - ❖ Research framework i.e., ecological, pilot, cross-sectional, or longitudinal etc.,
 - ❖ Sampling strategy
 - ❖ Who and what the target population was i.e., target sample size and group of focus (e.g., Adults only i.e., 18 years and above etc.,)

Description of Data [2]

[continue]

- **Describe the variables that's going to be used for the analysis:**
 - ❖ **Codebook (see next slide)** – provide a list of all the intended variables that is going to be analysed. You must mention specific details about variable – information such as the name, variable type (i.e., numeric or categorical) etc.,
 - ❖ **Initial sample characteristic table** – this is a breakdown on the number of observation documented for each variable (- i.e., what's present and missing).
 - ❑ Example 1, suppose the total is 100 and 89 respondents provided the ages. In this table, the mean age is calculated from that 89. You must report that the mean is based on 89 point, and 11 points are missing data.
 - ❑ Example 2, suppose the total is 100, where 51 were women and 42 were men. You must report the numbers and percentage for each category and report the numbers (& proportion) of missing data in the gender variable. This means including a third category: men (42), women (51), and unknown/missing (7)

A Video gamer's statistics



We have compiled the following information about the gaming habits of **Anwar Musah** (aka **The-PhD-Gamer**) across 3 console generations i.e., PlayStation 3, 4 and 5.

There are 161 game titles (~6,000 hours of game time) listed in the shared dataset (last updated 01/2024).

PSNProfiles (<https://psnprofiles.com/>)

Data Descriptor: [[Downloadable Dataset](#)]

This table is an example of codebook!

Variables Names	Descriptor
Number	[Numeric] Unique Identifier
GameTitle	[String] Name of the video game
Genre	[String] Type of genre (9 categories)
Platform	[String] Type of console (3 categories)
HourPlayed	[Numeric] Total number of hours invested in a game
CompletionRate	[Numeric] Percentage of completed content in a game
Status	[String] Current status of the game in terms of play is 'in-progress' or 'quit', or in 'hiatus' or 'done' as in completed (4 categories)
PlatinumTrophy	[Binary] 1 = 'Attained platinum trophy' and 0 = 'No platinum trophy'
DLCTrophies	[Binary] 1 = 'Yes' and 0 = 'No'. Are there any DLC trophies present (annoying feature as it effects the completion rate)?

We are going to use this data as a test dummy. We are just going to access RStudio server and then show how to import this dataset.

Then set you folks going for the practical material to prepare for the seminars on Thursday.

ID	Name	Age	Gender	Pathway	Gamer
0001	Brittany	18	Female	NA	No
0002	Idris	NA	Male	NA	NA
0003	Spike	19	Male	Political	Yes
0004	Lara	20	Female	Geography	Yes
0005	Nana-Kwesi	17	Male	Health	No
0006	Xiaoyu	23	Female	Geography	NA

Raw dataset

Characteristic	Breakdown on what’s present in the dataset
Age (in years)	Mean 19.7 (n = 5, 1 missing)
Gender	
Women	4 (66%)
Men	2 (33%)
Missing/Unknown	0 (0%)
Pathway	
Political	1 (16.5%)
Health	1 (16.5%)
Geography	2 (33%)
Missing/Unknown	2 (33%)
Gamer Status	
Yes	2 (33%)
No	2 (33%)
Missing/Unknown	2 (33%)

Show what’s available

To show missing numbers by category

```
table(dataset$pathway, useNA = “always”)
```

To exclude the missing entries

```
mean(dataset$age, na.rm = TRUE)
summary(dataset$age, na.rm = TRUE)
```

Description of Data [3]

[continue]:

Here, we make an explicit statement about missing data bias and complete case before proceeding with the analysis

- **Complete case analysis:** Here, you are making a declaration that you are sub-setting the dataset to cases who have complete information across **ALL** variables (no missing information), and that you are restricting the dataset & analysis to respondents without missing data.

ID	Name	Age	Gender	Pathway	Gamer
0001	Brittany	34	Female	NA	No
0002	Idris	NA	Male	NA	NA
0003	Spike	32	Male	Political	Yes
0004	Lara	29	Female	Geography	Yes
0005	Fiona	28	Female	Health	No
0006	Xiaoyu	29	Female	Geography	NA

Sample size = 6

Description of Data [4]

[continue]:

```
Code:  
dataset_nomissing <- dataset[complete.cases(dataset), ]
```

Here, we make an explicit statement about missing data bias and complete case before proceeding with the analysis

- **Complete case analysis:** Here, you are making a declaration that you are sub-setting the dataset to cases who have complete information across **ALL** variables (no missing information). You are restricting the dataset to respondents without missing data.

ID	Name	Age	Gender	Pathway	Gamer
0001	Brittany	34	Female	NA	No
0002	Idris	NA	Male	NA	NA
0003	Spike	32	Male	Political	Yes
0004	Lara	29	Female	Geography	Yes
0005	Fiona	28	Female	Health	No
0006	Xiaoyu	20	Female	Geography	NA

Sample size = 6

Delete 3 rows

Description of Data [5]

[continue]:

```
Code:  
dataset_nomissing <- dataset[complete.cases(dataset), ]
```

Here, we make an explicit statement about missing data bias and complete case before proceeding with the analysis

- **Complete case analysis:** Here, you are making a declaration that you are sub-setting the dataset to cases who have complete information across **ALL** variables (no missing information). You are restricting the dataset to respondents without missing data and use it for the analysis.

ID	Name	Age	Gender	Pathway	Gamer
0003	Spike	32	Male	Political	Yes
0004	Lara	29	Female	Geography	Yes
0005	Fiona	28	Female	Health	No

Sample size = 6
Delete 3 rows

Retained sample = 3
Drop-out rate = 50%

- **Missing data bias:** Here, you must acknowledge that due to the sample reducing from what was initially A to B, and restricting it to complete case sample, you have introduced some bias in your results.

Let's make a mental summary of what we've learnt in the
past 4 weeks

Thank you

